

# ØKA2015 Applied Data Science

University of Inland Norway, Lillehammer, autumn 2026

## Assignment 1

Reading and inspecting data with `readr` and `dplyr`

**Due** **Wednesday 30 September 2026, 23:59**, on Canvas  
**Submission** individual; see Section 4  
**Status** one of four compulsory assignments; three of the four must be approved to sit the exam

### 1. What this assignment is

This assignment practises the first steps of every data project: getting a file into R, looking at it properly, and making a first change to it. It has **eight problems**. Six are answered with a few lines of R code; two are answered in writing.

Everything you need is covered in Lectures 2, 3 and 4:

| Problems | What they need                                                                                                                                                     | Covered in          |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 1 to 4   | <code>read_csv()</code> , <code>head()</code> , <code>ls()</code> , <code>nrow()</code> , <code>ncol()</code> ,<br><code>glimpse()</code> , <code>summary()</code> | Lecture 3 (16 Sept) |
| 5 to 6   | <code>select()</code> , <code>mutate()</code> , the pipe <code> &gt;</code>                                                                                        | Lecture 4 (18 Sept) |
| 7 to 8   | what <code>dplyr</code> is for; what data wrangling is                                                                                                             | Lectures 2 and 4    |

Start Problems 1 to 4 right after Lecture 3. They are the same steps as the Data Lab, applied to a different file.

### 2. The data

**California Housing Prices:** one row per district in California from the 1990 census. It is the dataset distributed on Kaggle ([camnugent/california-housing-prices](https://www.kaggle.com/camnugent/california-housing-prices)). A copy is hosted in the course repository, so you can read it in one line, with no account and no download:

```
library(tidyverse)
housing <- read_csv("https://raw.githubusercontent.com/touseefhameed/oka2015-applied-data-science/main/data/housing.csv")
```

If you prefer a local file, download it from that address or from Kaggle, save it as `data/housing.csv` inside your project folder, and use `read_csv("data/housing.csv")`.

| Column                                         | Meaning                                              |
|------------------------------------------------|------------------------------------------------------|
| <code>longitude</code> , <code>latitude</code> | location of the district                             |
| <code>housing_median_age</code>                | median age of the houses, in years                   |
| <code>total_rooms</code>                       | total number of rooms in the district                |
| <code>total_bedrooms</code>                    | total number of bedrooms (a few values are missing)  |
| <code>population</code>                        | number of people living in the district              |
| <code>households</code>                        | number of households in the district                 |
| <code>median_income</code>                     | median household income, in tens of thousands of USD |
| <code>median_house_value</code>                | median house value, in USD                           |
| <code>ocean_proximity</code>                   | how close the district is to the ocean (text)        |

### 3. Problems

For every problem, show the code you ran *and* the output it produced. Write a comment above or beside each command saying what it does.

**Problem 1. Loading the data.** Load the tidyverse, read the dataset into an object called `housing` with `read_csv()`, and display its first 6 rows.

**Problem 2. Your environment.** Run `ls()`. Which objects does your R session contain now, and which one holds the data?

**Problem 3. Size.** How many observations and how many variables does the dataset have? Use one function for each, and state the two numbers in a sentence.

**Problem 4. Structure and summary.** Show the structure of the dataset with `glimpse()` and its basic statistics with `summary()`. Then answer in one or two sentences: which column contains text rather than numbers, and which column has missing values?

**Problem 5. Selecting columns.** Keep only `median_house_value`, `housing_median_age`, `total_rooms` and `median_income`, and display the first 10 rows of the result.

**Problem 6. Creating a column.** Add a new column `rooms_per_household`, equal to `total_rooms` divided by `households`, and display the first 10 rows of the updated dataset.

**Problem 7. Written, about 100 words.** What is `dplyr` for? Name five of its functions and say, in one line each, what they do.

**Problem 8. Written, about 100 words.** What is data wrangling, and why does it take such a large share of a data scientist's time? Use an example from this dataset.

## 4. What to hand in

Students should utilize R Markdown for proper documentation: an `.Rmd` file with the eight problems as headings, the code and its output under each, and the written answers as text, knitted to PDF or HTML. If R Markdown does not work on your computer, screenshots of your RStudio window showing each command and its output are accepted instead.

Upload to Canvas, named with your own name:

1. `YourName_Assignment1.R`: a script containing every command you used, with a comment for each one. It must run from top to bottom in a fresh R session.
2. `YourName_Assignment1.Rmd` together with its knitted `YourName_Assignment1.pdf` (or `.html`); or, if knitting fails, screenshots of your RStudio window with the code and its output, plus the written answers to Problems 7 and 8 in a document.

## 5. How it is assessed

The assignment is marked **approved** or **not approved**. It is approved when:

- the script runs without errors in a fresh session, and each command has a comment;
- the R Markdown document, or the screenshots, show both the code and its output for Problems 1 to 6;
- the written answers to Problems 7 and 8 are in your own words.

## 6. Working with others, and AI tools

Discussing the assignment with other students is encouraged. Submitting someone else's code or text as your own is plagiarism and is handled under the university's rules on academic honesty.

The use of AI tools is **not recommended** for this assignment. If you do use one, follow the university's guidelines on artificial intelligence in studies and examinations, do not use it to produce the answers, and state at the end of your submission what you used it for.

## 7. Help

- The materials for Lectures 1 to 4: the slides on Canvas, and the scripts, tutorials and practice exercises in the course repository, <https://github.com/touseefhameed/oka2015-applied-data-science>
- *R for Data Science* (2e), free online: Ch. 2 *Workflow: basics*, Ch. 7 *Data import*, Ch. 3 *Data transformation*, <https://r4ds.hadley.nz>
- Questions: ask me in class, or email [touseef.hameed@inn.no](mailto:touseef.hameed@inn.no).

Data: Pace, R. K. and Barry, R. (1997), from the StatLib repository; the CSV form is the one distributed on Kaggle and in Géron, *Hands-On Machine Learning*. Used for teaching only.